Weight of Evidence, Statistical Power, and Posterior Odds

Weight of Evidence

Suppose we want to compare two hypotheses, H_0 and H_1, after observing some data D.

A natural measure of the weight of evidence supplied by the data is the likelihood ratio:

$$LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)}$$

The likelihood ratio tells us how much more likely the observation is under H_1 than under H_0.

For example,

$$LR_{10} = 4$$

means that the observation is four times more likely under H_1 than under H_0.

Thus, the data provide 4:1 evidence in favor of H_1 over H_0.

Let's begin with a simple example.

Example 1: Two heads in two coin flips

Suppose we have two hypotheses about a coin:

- H_0: the coin is fair
- H_1: the coin is tricked and has two heads

Under H_0,

$$P(\text{head} \mid H_0) = \frac{1}{2}$$

Under H_1,

$$P(\text{head} \mid H_1) = 1$$

Now flip the coin twice and observe two heads

$$D = 2 \text{ heads}$$

Under H_0 (and assuming independence),

$$P(D \mid H_0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 0.25$$

Under H_1,

$$P(D \mid H_1) = 1$$

Therefore, the likelihood ratio is

$$LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)} = \frac{1}{0.25} = 4$$

Thus,

$$\boxed{LR_{10} = 4}$$

Observing two heads is four times more likely under H_1 than under H_0.

The observation therefore provides 4:1 evidence in favor of H_1.

From weight of evidence to posterior odds

The likelihood ratio tells us how strongly the observation favors one hypothesis over the other. But suppose we want to answer a different question:

> After observing the data, what are the odds of H_1 relative to H_0?

To calculate these posterior odds, we also need to specify the prior odds.

Suppose the prior odds are

$$\boxed{H_1 : H_0 = 1 : 9}$$

Posterior odds are obtained by multiplying the prior odds by the likelihood ratio:

$$\text{posterior odds} = \text{likelihood ratio} \times \text{prior odds}$$

Therefore,

$$\text{posterior odds} = 4 \times \frac{1}{9} = \frac{4}{9}$$

Thus, the posterior odds are

$$\boxed{H_1 : H_0 = 4 : 9}$$

To convert posterior odds into a posterior probability,

$$P(H_1 \mid D) = \frac{\frac{4}{9}}{1 + \frac{4}{9}} = \frac{4}{13} \approx 0.31$$

Thus,

$$\boxed{P(H_1 \mid D) \approx 0.31}$$

Similarly,

$$P(H_0 \mid D) = \frac{\frac{9}{4}}{1 + \frac{9}{4}} = \frac{9}{13} \approx 0.69$$

Therefore,

$$\boxed{P(H_0 \mid D) \approx 0.69}$$

Notice what happened:

$$\boxed{1 : 9 \xrightarrow{LR=4} 4 : 9}$$

The observation favors H_1 over H_0, because $LR_{10} > 1$.

But the evidence is not strong enough to overcome the initial 9:1 odds in favor of H_0.

Example 2: Statistical significance in a low-powered study

Now consider a conventional null-hypothesis significance test.

Suppose the study report a statistically significant result ($\alpha = 0.05$):

$$D = p < 0.05$$

By definition,

$$P(D \mid H_0) = 0.05$$

Then, suppose the study has **20% statistical power** ($\beta = 0.8$) representative of the low levels of statistical power reported in many neuroscientific studies.

Power is the probability of obtaining a statistically significant result when H_1 is true:

$$P(D \mid H_1) = 1 - \beta$$

For 20% power,

$$P(D \mid H_1) = 0.20$$

Weight of evidence

The likelihood ratio associated with the observation is

$$LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)}$$

Therefore,

$$LR_{10} = \frac{0.20}{0.05} = 4$$

Thus,

$$\boxed{LR_{10} = 4}$$

A statistically significant result is four times more likely under H_1 than under H_0.

The significant result therefore provides 4:1 evidence in favor of H_1.

Notice that this is exactly the same weight of evidence as observing two heads in the coin flip example:

$$\boxed{LR_{10}^{\text{coin flip}} = LR_{10}^{\text{20\% power}} = 4}$$

Knowing only that the study produced $p < 0.05$, the weight of evidence is the same in these two examples.

Posterior odds

Again suppose our prior odds are

$$\boxed{H_1 : H_0 = 1 : 9}$$

The posterior odds are

$$\text{posterior odds} = 4 \times \frac{1}{9} = \frac{4}{9}$$

Therefore,

$$\boxed{H_1 : H_0 = 4 : 9}$$

Just as in the coin flip example,

$$\boxed{1 : 9 \xrightarrow{LR=4} 4 : 9}$$

Converting these posterior odds into a posterior probability,

$$P(H_1 \mid D) = \frac{\frac{4}{9}}{1 + \frac{4}{9}} = \frac{4}{13} \approx 0.31$$

Thus,

$$\boxed{P(H_1 \mid p < 0.05) \approx 0.31}$$

Likewise,

$$P(H_0 \mid D) = \frac{\frac{9}{4}}{1 + \frac{9}{4}} = \frac{9}{13} \approx 0.69$$

Therefore,

$$\boxed{P(H_0 \mid p < 0.05) \approx 0.69}$$

The statistically significant result provides evidence in favor of H_1, but that evidence is not sufficient to overcome prior odds of 9:1 in favor of H_0.

What does $p < 0.05$ tell us?

The significance threshold tells us that

$$P(p < 0.05 \mid H_0) = 0.05$$

It does not tell us the posterior odds of H_1 versus H_0.

To calculate posterior odds, we need three quantities:

$$P(p < 0.05 \mid H_0) \qquad P(p < 0.05 \mid H_1) \qquad \frac{P(H_1)}{P(H_0)}$$

The first two determine the likelihood ratio:

$$LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)}$$

The last one is the prior odds.

We then combine the likelihood ratio with the prior odds to obtain the posterior odds.

Example 3: Statistical significance in an 80%-powered study

Now consider the same statistical test, but suppose the study has 80% power.

Therefore,

$$P(D \mid H_0) = 0.05$$

But with 80% power,

$$P(D \mid H_1) = 0.80$$

Weight of evidence

The likelihood ratio is now

$$LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)} = \frac{0.80}{0.05} = 16$$

Thus,

$$\boxed{LR_{10} = 16}$$

A significant result is now 16 times more likely under H_1 than under H_0.

The observation therefore provides 16:1 evidence in favor of H_1.

Compare this with the low-powered study:

$$\begin{aligned} 20\% \text{ power}: &\quad LR_{10} = 4 \\ 80\% \text{ power}: &\quad LR_{10} = 16 \end{aligned}$$

The same criterion,

$$p < 0.05$$

can therefore correspond to very different weights of evidence depending on the power of the experiment.

Posterior odds

Keep the prior odds at

$$H_1 : H_0 = 1 : 9$$

The posterior odds are now

$$\text{posterior odds} = 16 \times \frac{1}{9} = \frac{16}{9}$$

Therefore,

$$\boxed{H_1 : H_0 = 16 : 9}$$

The updating is now

$$\boxed{1 : 9 \xrightarrow{LR=16} 16 : 9}$$

Converting these posterior odds into a posterior probability,

$$P(H_1 \mid D) = \frac{\frac{16}{9}}{1 + \frac{16}{9}} = \frac{16}{25} = 0.64$$

Thus,

$$\boxed{P(H_1 \mid p < 0.05) = 0.64}$$

Similarly,

$$P(H_0 \mid D) = \frac{\frac{9}{16}}{1 + \frac{9}{16}} = \frac{9}{25} = 0.36$$

Therefore,

$$\boxed{P(H_0 \mid p < 0.05) = 0.36}$$

Unlike the 20%-powered study, the evidence is now strong enough to overcome the initial 9:1 odds in favor of H_0.

Comparing the three examples

In all three examples, we use the same prior odds:

$$H_1 : H_0 = 1 : 9$$

What changes is the weight of evidence supplied by the observation.

Observation	$P(D \mid H_0)$	$P(D \mid H_1)$	Weight of evidence LR_{10}	Prior odds $H_1 : H_0$	Posterior odds $H_1 : H_0$	$P(H_1 \mid D)$	$P(H_0 \mid D)$
Two heads in two flips	0.25	1.00	**4**	1:9	**4:9**	**0.31**	**0.69**
$p < 0.05$, 20% power	0.05	0.20	**4**	1:9	**4:9**	**0.31**	**0.69**
$p < 0.05$, 80% power	0.05	0.80	**16**	1:9	**16:9**	**0.64**	**0.36**

The logic is identical in each case.

First, calculate the weight of evidence:

$$\boxed{LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)}}$$

Then, if we want posterior odds, combine the weight of evidence with the prior odds:

$$\boxed{\text{posterior odds} = LR_{10} \times \text{prior odds}}$$

Finally, convert posterior odds to posterior probabilities.

If the posterior odds are $a : b$, then

$$P(H_1 \mid D) = \frac{a}{a+b}$$

and

$$P(H_0 \mid D) = \frac{b}{a+b}$$

Take-home message

The **weight of evidence** supplied by an observation is quantified by the likelihood ratio:

$$\boxed{LR_{10} = \frac{P(D \mid H_1)}{P(D \mid H_0)}}$$

For a statistically significant result considered simply as the binary event

$$D = p < \alpha$$

this becomes

$$\boxed{LR_{10} = \frac{1-\beta}{\alpha}}$$

Thus, with $\alpha = 0.05$,

$$20\% \text{ power} \quad \Longrightarrow \quad LR_{10} = 4$$

whereas

$$80\% \text{ power} \quad \Longrightarrow \quad LR_{10} = 16$$

If we then want to determine the **posterior odds**, the weight of evidence alone is not enough. We also need the prior odds:

$$\boxed{\text{posterior odds} = \text{weight of evidence} \times \text{prior odds}}$$

With prior odds of $1 : 9$,

$$\boxed{1 : 9 \xrightarrow{LR=4} 4 : 9}$$

for both the coin flip example and the 20%-powered study, corresponding to

$$\boxed{P(H_1 \mid D) = \frac{4}{13} \approx 0.31}$$

For the 80%-powered study,

$$\boxed{1 : 9 \xrightarrow{LR=16} 16 : 9}$$

corresponding to

$$\boxed{P(H_1 \mid D) = \frac{16}{25} = 0.64}$$

The likelihood ratio tells us how much the evidence changes the odds.

The prior odds tell us where we start.

The posterior odds tell us where we end up.

The posterior probabilities convert those final odds into probabilities.